Yen-Nan (Aaron) Ho

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AI Engineer

Data Scientist at Harbor Tech, developing two flagship products under Harbor AI 2.0: an Industrial Intelligent Monitoring Platform for ongoing equipment monitoring and diagnostics, and an Equipment End-of-Line Quality Inspection Platform for customer-specific unit-by-unit inspection. Led monitoring product/domain design and the inspection project from requirements and specifications through implementation, UI/UX, and on-site delivery. Built traceable issue/project records and parallel AI-assisted development, repair, and verification workflows, supported by daily functional regression and human release review.

Core Skills

- **Industrial AI/Research:** remaining useful life (RUL), phase-resolved partial discharge (PRPD)/vibration, FFT, anomaly detection, CNNs, ablations, confidence, and data-availability assessment.
- **AI Systems:** LangChain/LangGraph, report services, vLLM, OpenAI-compatible APIs, deterministic facts, fallbacks, and human validation.
- **Platform/Delivery:** Python, Django/Django Ninja, PostgreSQL, Redis, RabbitMQ, REST, async workflows, React/TypeScript, Docker, offline deployment, update/rollback, Git, and automated tests.

Work Experience

Data Scientist, Harbor Tech | Taipei | Oct 2025-Present

- **Harbor AI 2.0 Industrial Intelligent Monitoring Platform:** Led product/domain design and built monitoring, diagnostics, and reporting capabilities from scratch with the team. Managed legacy/new-system coexistence and data migration, progressively replacing the legacy system; 2.0 now operates without dependence on 1.0.
- **Harbor AI 2.0 Equipment End-of-Line Quality Inspection Platform:** Led requirements, specifications, AI-assisted implementation, and on-site delivery with engineering partners. Reorganized a spec-compliant UI requiring five to six page levels into three core interface types: plan list, work order, and board. Separated installation confirmation from measurement, adding mandatory sensor-ID/diagram checks. Demonstrated the system, trained users, and collected feedback on October 12, 2026; formal acceptance remains unconfirmed.
- **AI development and project records:** Built traceable issue/project records linking problems, analysis, decisions, results, verification, and versions. Organized AI agents for parallel development, repair, and verification on isolated branches. Ran daily automated functional regression and monitoring, routing findings to issues and Bitbucket PRs. I define requirements, quality criteria, and domain decisions; reviewers and I decide merge and release.
- **Model assumptions and validation:** Audited AI-generated RUL conclusions. Traced the data source, found test constants misrepresented as observations, and invalidated the conclusion. Then required multi-parameter, multi-curve, and forward/reverse ablations to recheck the assumptions. Connected research notebooks to production code to reduce drift. Separated fault type, severity, RUL, and low-confidence states.

- **Report-service integration:** Turned field-analysis and manually reviewed reporting methods into reusable workflows. Used a LangGraph POC to separate analysis results from LLM narrative, then advanced the platform-connected report service; DOCX reports and frontend downloads were integrated into the mainline. Separately validated live vLLM endpoint capabilities.
- **Platform and delivery:** Collaborated on Django async workflows, defining APIs, repository/artifact-by-reference contracts, task lifecycles, and validation boundaries. Integrated Docker/offline and cross-architecture installation, platform update/rollback, and auditable migration; led release integration and cut decisions.

Data Science Intern, Institute of Sociology, Academia Sinica | Taipei | May 2019-Dec 2020

Used survey-area grouping, point-to-point distances, OR-Tools TSP, and Folium for field-survey routes, dispersion, and budget estimation.

Selected Research and Projects

- **MCIENet, M.S. thesis (2024):** Independently led research decomposition, multi-scale CNNs, data pipelines, ablations, cross-dataset evaluation, and explainable AI (XAI) for DNA-sequence-based 3D chromatin interaction prediction. Adapted parts of ChINN preprocessing; retained cross-cell-line generalization limits.
- **Horus, undergraduate capstone (2020):** Team-integrated multi-camera tracking, person ReID, and face recognition. Primarily owned ReID integration, similarity/dual-threshold decisions, and tracking-record workflows. The team won first place in the undergraduate capstone competition.
- **Taipei Disaster Resource Dashboard (2025):** Proposed the topic and collected data for a team project on disaster-prevention resources around tourist attractions. Finalist, Taipei-New Taipei City Dashboard Hackathon; the public dashboard is based on a fork of a friend' s template.
- **PropGMAN (2022):** Proposed the direction, led a three-person team, and handled model implementation, experiments, and the final-round report using third-party GMAN for regional land indicators. Team Creative Award, AI for Asset Activation Innovation Contest.
- **Financial Recommendation and Notifications (2020):** Proposed the topic and led team coordination. Primarily designed, implemented, and deployed the Telegram chatbot, integrated team modules, and advised on recommendation algorithms. Team Excellence Award, financial-programming course competition.

Research and Publications

- Dynamic traffic network representation model for improving the prediction performance of passenger flow. Knowledge-Based Systems, 2024. Supporting code and technical assistance.
- A Dynamic Embedding Method for Passenger Flow Estimation. IIAI AAI, 2021. Method ideation, data processing, model construction, and experiments. Competitive Paper Award.
- Hierarchical Attention Network with Sentiment Lexicons for Stock Article Emotion Analysis. TCSE, 2020. Performed all research work; coauthored publication.

Education

- National Chengchi University, M.S. in Computer Science | Sep 2021-Sep 2024.
- Soochow University, B.S. in Data Science | Sep 2017-Jul 2021.